

PreSCAN: Fast NeRF Architecture Selection for Satellite Scene Reconstruction

Devjyoti Chakraborty
University of Georgia
Athens, GA, USA
devjyoti.chakraborty@uga.edu

Kriti Ghosh
University of Georgia
Athens, GA, USA
kriti.ghosh@uga.edu

Lakshmish Ramaswamy
University of Georgia
Athens, GA, USA
laksmr@uga.edu

Zaki Sukma
University of Georgia
Athens, GA, USA
zaki.sukma@uga.edu

In Kee Kim
University of Georgia
Athens, GA, USA
inkee.kim@uga.edu

Nancy K. O'Hare
University of Georgia
Athens, GA, USA
nkohare@uga.edu

Rakandhiya D. Rachmanto
University of Georgia
Athens, GA, USA
rakandhiya.rachmanto@uga.edu

Suchendra M. Bhandarkar
University of Georgia
Athens, GA, USA
suchi@uga.edu

Deepak Mishra
University of Georgia
Athens, GA, USA
dmishra@uga.edu

Abstract

Operational satellite reconstruction pipelines build large-scale 3D models from multi-date, multiview imagery. Even with conventional photogrammetric methods, reconstructing each area of interest takes hours of compute, and a full region requires repeating this across many areas of interest at regional or national scale. Neural Radiance Fields (NeRF) have recently emerged as a high-fidelity alternative for this task, but deploying them at this scale is difficult. Each scene requires individual training, and selecting an architecture through Neural Architecture Search (NAS) takes hours of GPU time per scene, which is prohibitive for large-scale workflows.

Our SHAP analysis across 73 satellite scenes and 10,512 trained NeRF configurations reveals that multi-view consistency, rather than model architecture, determines reconstruction quality. Based on this insight, we develop PreSCAN, a predictive framework that estimates reconstruction quality prior to training using lightweight geometric and photometric descriptors. PreSCAN selects architectures comparable to NAS-chosen ones in under 30 seconds, a 1000× speedup, with < 1 dB PSNR prediction error. We further present a deployment case study on edge platforms (Jetson Orin), where combining its predictions with offline cost profiling reduces inference power by 26% and latency by 43% with minimal quality loss. Experiments on the DFC2019 dataset across suburban (Jacksonville) and industrial (Omaha) scenes confirm that PreSCAN maintains accuracy across these contrasting scene types without retraining, making it a practical, deployment-ready screening tool for operational satellite reconstruction pipelines.

CCS Concepts

• **Information systems** → **Geographic information systems**; • **Computing methodologies** → **Computer vision**; **Neural networks**; *Shape modeling*.

Keywords

Neural Radiance Fields, Satellite Imagery, Remote Sensing, 3D Reconstruction, Architecture Selection, Edge Computing

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1 Introduction

Operational satellite reconstruction systems build large-scale 3D models of urban areas from multi-date, multiview imagery. Even with conventional photogrammetric pipelines, reconstructing each area of interest takes hours of compute [14]. As a result, reconstructing a full region requires repeating the pipeline across many areas of interest. The number of such areas is large, often covering regional or national scales [3].

Neural Radiance Fields (NeRF) have recently emerged as a high-fidelity alternative for this reconstruction task, enabling novel-view synthesis (NVS) from multi-view images [24]. Satellite-specific variants adapt NeRF to overhead observation, addressing challenges such as RPC camera models, multi-date acquisitions, illumination shifts, and transient objects [16, 21, 32]. These variants promise higher-fidelity reconstruction, but they increase the per-scene cost even further, as each scene now also requires its own architecture search and training [4, 31].

Recent advances have focused on accelerating NeRF training through faster optimizers, compact networks, and improved sampling [2, 10, 15, 17, 26]. While these reduce training time, they



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do not eliminate the need for scene-specific optimization nor enable cross-scene generalization. We empirically observe that NAS [23, 40] requires roughly ten hours of GPU time per scene on A6000 GPUs (Table 2, §4), which is prohibitively expensive for satellite workflows that process many locations and scenes. Critically, these approaches prioritize architectural optimization while overlooking the fundamental issue that input view consistency, not model architecture, determines NeRF reconstruction quality in satellite imagery. These approaches are built on a common assumption that architectural optimization is essential for high-quality reconstruction. Our work begins by examining the validity of this assumption.

Through extensive SHAP analysis across 73 satellite scenes and 10,512 trained NeRF configurations, we discover that multi-view consistency, rather than model architecture, determines reconstruction quality. Specifically, scene-level descriptors (e.g., photometric variance, view angle alignment, geometric coverage) consistently outweigh architectural parameters, including network depth, width, and sampling density. This finding suggests that the computational overhead of per-scene NAS can be avoided by predicting reconstruction quality from scene characteristics alone.

Based on this insight, we present PreSCAN, a predictive framework that estimates NeRF reconstruction quality before training, eliminating costly per-scene NAS. PreSCAN jointly encodes architectural parameters and scene-level descriptors (e.g., view alignment, photometric variance, and spatial coverage) to predict reconstruction quality within seconds. By leveraging both feature types, PreSCAN achieves strong generalization with minimal training data without requiring expensive architectural search.

PreSCAN is designed to integrate into operational remote sensing pipelines as a lightweight triage stage between image acquisition and reconstruction. Our evaluation spans diverse scenes with varying terrain, lighting, and view configurations. Tests on DFC2019 datasets [34] from Jacksonville (suburban) and Omaha (industrial) demonstrate that PreSCAN maintains prediction accuracy across these contrasting scene types without retraining.

PreSCAN selects architectures comparable to NAS-chosen ones in under 30 seconds, a 1000× speedup over NAS, while keeping PSNR prediction error under 1 dB. It maintains accuracy across contrasting scene types without retraining and remains robust under limited supervision. Beyond the architecture selection, we show that combining its predictions with offline cost profiling makes high-fidelity reconstruction practical on power- and latency-constrained edge platforms such as Jetson Orin. These results position PreSCAN as a practical, deployment-ready screening tool for operational satellite reconstruction pipelines.

Contributions. Our key contributions are as follows:

- (i) **[Analysis]** Through SHAP analysis over 73 satellite scenes and 10,512 NeRF configurations, we show that multi-view consistency, not model architecture, determines satellite NeRF reconstruction quality.
- (ii) **[Method]** We develop PreSCAN, a predictive framework that predicts reconstruction quality before training and selects architectures in under 30 seconds, a 1000× speedup over NAS.
- (iii) **[Deployment]** We show on edge platforms (Jetson Orin) that combining PreSCAN with offline cost profiling cuts inference power by 26% and latency by 43% with < 1 dB quality loss.

2 Background and Related Work

2.1 NeRF and Its Variants for Satellite Imagery

NeRF [24] generate photorealistic novel views by learning a continuous volumetric scene function from multi-view color images. A neural network maps a 3D position $\mathbf{x}$ and viewing direction $\mathbf{d} \in \mathbb{R}^3$ to color and volume density, $F_\theta(\mathbf{x}, \mathbf{d}) \rightarrow (\mathbf{c}, \sigma)$.

The final rendered color along a ray r is computed as:

$$\mathbf{c}(r) = \sum_{i=1}^N T_i \alpha_i \mathbf{c}_i, \quad (1)$$

with $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$ and $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$. This formulation allows NeRF to synthesize view-consistent appearance and geometry, but it requires per-scene training, limiting its scalability in practice.

Satellite imagery poses unique challenges for NeRF: dynamic elements, long capture intervals, illumination shifts [7, 9, 42], and non-pinhole RPC camera models [11], all introducing view inconsistencies that often degrade reconstruction quality.

Shadow NeRF (S-NeRF) [8] extends NeRF for satellite imagery by incorporating solar angle ω into a shadow-aware illumination:

$$\mathbf{c}(\mathbf{x}, \omega) = \mathbf{c}_a(\mathbf{x}) \cdot (s(\mathbf{x}, \omega) + (1 - s(\mathbf{x}, \omega)) \cdot \mathbf{a}(\omega)) \quad (2)$$

where $\mathbf{c}_a(\mathbf{x})$ is the albedo, $s(\cdot)$ is a predicted shadow, and $\mathbf{a}(\cdot)$ is ambient lighting from the sun. By decoupling geometry and lighting, S-NeRF improves robustness under illumination variation. We adopt S-NeRF as our base model for its satellite-specific design and efficiency. We also use SatNeRF [21] combined with S-NeRF for testing model invariance. SatNeRF defines $F : (\mathbf{x}, \omega, t_j) \rightarrow (\sigma, \mathbf{c}_a, s, \mathbf{a}, \beta)$. Relative to S-NeRF, it adds ambient illumination $\mathbf{a}$ and a transient uncertainty term β , with an additional input dependency of t_j , time embeddings attributed to each image.

2.2 Related Work

Generalizable Neural Radiance Fields. NeRF models typically require scene-specific training and struggle to generalize across environments [22, 44]. Recent work has proposed generalizable NeRFs [5, 12, 18, 37, 43] that improve robustness to sparse inputs, varying illumination, and geometric variation. However, most methods still require retraining per scene, and they do not address expensive architectural search overhead.

WaveNeRF [43] uses wavelet-based decompositions to preserve fine details with few input views, while MVSNeRF [5] introduces cost volume aggregation for sparse inputs. Geometry-aware methods like GPSNeRF [12] and ACA-based NeRFs [18] fuse geometric, photometric, and sometimes semantic information to improve reconstruction. NeRF Transformer models [35] and Gen-NeRF [5] emphasize hardware efficiency or architectural design. Despite these advances, these methods still rely on scene-specific training and do not support fast architecture selection.

Neural Architecture Search (NAS) for NeRF. NAS has been applied to reduce NeRF training cost by designing efficient, scene-specific models [1, 13, 23, 27, 30, 33, 40]. NAS-NeRF [27], for example, uses a generator-inquisitor framework to search over modular MLP architectures. While such approaches reduce training time,

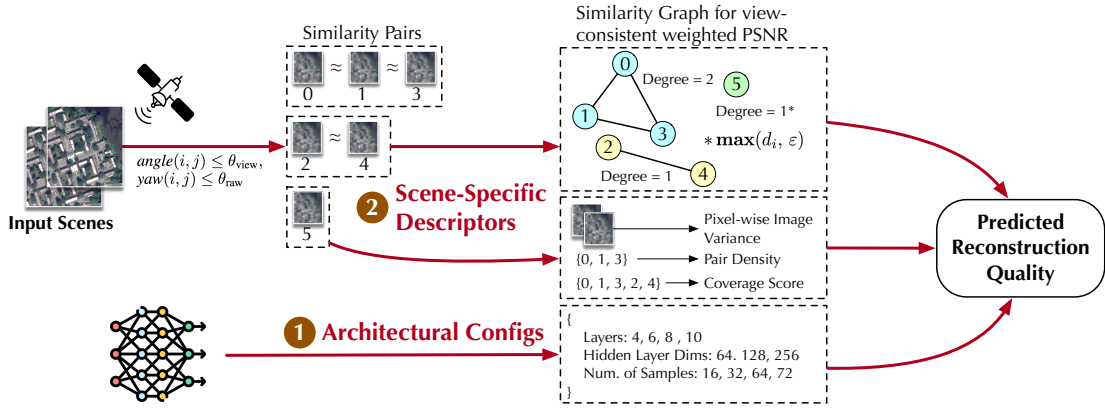


Figure 1: Overview of the PreSCAN framework. PreSCAN predicts suitable NeRF architectures before training by modeling the interaction between architectural parameters and scene descriptors. Using lightweight inputs (e.g., view alignment, sampling patterns), it enables fast (< 30 sec.), scalable architecture selection without per-scene retraining.

they still require training multiple candidate models per scene, limiting scalability.

In contrast, PreSCAN avoids per-scene training entirely by predicting reconstruction quality using lightweight scene-level and architectural descriptors. This enables fast screening of candidate architectures, reducing NeRF search time from hours to < 30 seconds per scene.

3 The PreSCAN Framework

3.1 Scene-Aware NeRF Architecture Prediction

Our goal is to identify suitable NeRF architectures for satellite scenes *prior to any training*, avoiding the costly process of repeatedly training and evaluating models per scene. PreSCAN addresses this by estimating NeRF reconstruction performance using a combination of architectural configurations and scene-level descriptors. This approach removes iterative training loops and enables rapid, scalable architecture selection across satellite scenes.

NeRF relies on local ray-based sampling to infer geometry and appearance, making it more sensitive to geometric and photometric consistency than to high-level semantics [6, 25, 38]. Satellite imagery complicates this process due to shadows, temporal gaps, and transient objects like clouds or vehicles. To address these challenges, PreSCAN models the interaction between scene-level complexity and architectural design to predict their impact on NeRF reconstruction quality.

Figure 1 illustrates the PreSCAN workflow. The model takes two input types: (i) *architectural configurations* (e.g., number of layers, neurons, ray samples), and (ii) *scene-specific descriptors* that capture photometric variance, ray alignment, and spatial coverage across views. The two input sets are combined in a lightweight regression model that learns how architectural capacity interacts with scene complexity to influence performance. Unlike NAS-based approaches that require full model training for each configuration, PreSCAN enables *fast, interpretable architecture screening* with minimal computational overhead.

3.2 Feature Design and SHAP-based Analysis

The effectiveness of PreSCAN relies on input features that accurately reflect the factors influencing NeRF performance. Unlike traditional vision tasks that benefit from semantic embeddings or high-level abstractions [19, 36, 41], NeRF reconstruction is more sensitive to pixel-level geometry and color consistency across multiple views. As a result, NeRF is especially sensitive to variations in viewpoint or appearance, even when the overall scene content appears structurally similar.

We design geometric and photometric descriptors that capture scene consistency across views, including inverse PSNR, view angle and intensity variance, and cosine similarity. These features quantify multi-view coherence and indicate a scene’s suitability for NeRF reconstruction.

To assess feature importance, we perform SHAP analysis [20] using multi-view satellite scenes from the DFC2019 dataset [34]. We analyze two contrasting regions: Jacksonville (JAX), a suburban area, and Omaha (OMA), an urban industrial zone with distinct terrain and view distributions.

As shown in Figure 2a and 2b, SHAP values indicate that inverse PSNR (w'), photometric variance ($\text{Var}(I_i)$), and view angle cosine similarity ($\cos \text{sim}$) are the most important predictors across both datasets. In contrast, architectural parameters, e.g., the number of layers or ray samples, have minimal influence. This suggests that input consistency plays a more critical role in reconstruction quality than architectural depth or complexity. We validate this finding by applying SHAP analysis to a combined dataset of JAX and OMA scenes (Figure 2c). The same three features remain dominant, suggesting that the model captures general patterns relevant to reconstruction, rather than overfitting to dataset-specific artifacts.

In summary, consistency-based features, e.g., inverse PSNR, photometric variance, and cosine similarity, are the most influential for predicting NeRF performance. These descriptors remain consistently important across the scenes we analyze.

Based on this analysis, we hypothesize and validate that **pixel-level consistency – both geometric and photometric – across nearby views is the key driver of NeRF reconstruction quality**

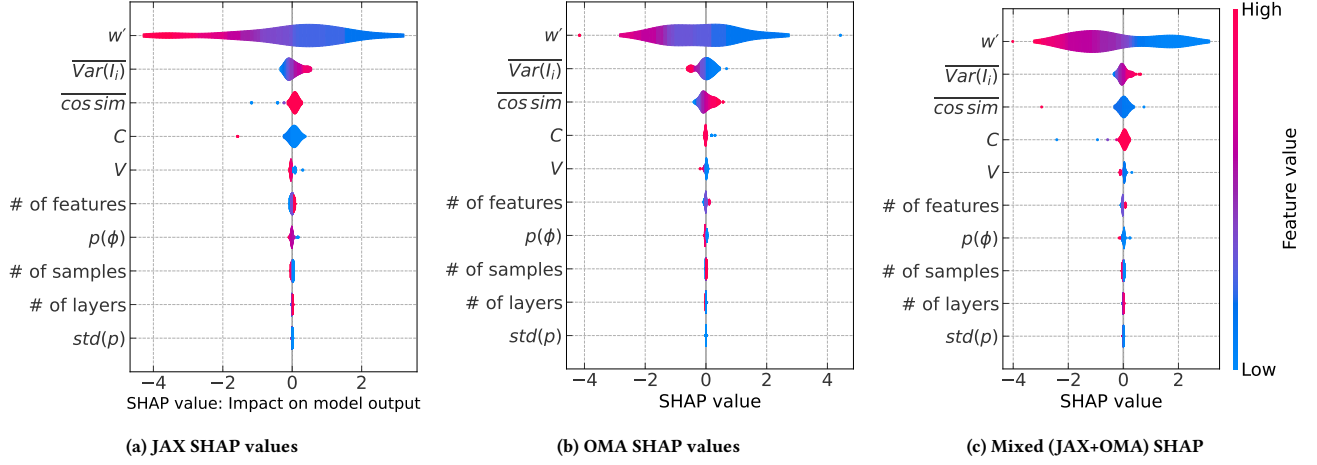


Figure 2: SHAP value analysis across JAX, OMA, and combined datasets. (a) ‘JAX scenes’: Inverse PSNR (w'), photometric variance ($\text{Var}(I_i)$), and cosine similarity (overline cos sim) are the most influential. (b) ‘OMA scenes’: The same top features are consistently identified, despite scene differences. (c) ‘Combined dataset’: Feature rankings remain stable, confirming their general importance across datasets.

over model specific features. This hypothesis guides our evaluations, where we assess prediction accuracy under varying scene conditions.

3.3 Predictive Feature Representation

We develop a lightweight feature representation that jointly encodes (i) *architectural configurations* and (ii) *scene-specific descriptors* (or scene-level geometric and radiometric features) to estimate NeRF reconstruction quality before training. This allows fast and interpretable performance estimation without requiring rendering or optimization.

(i) Architectural Configurations. We use four parameters (descriptors) for each NeRF architecture: number of MLP layers (L), neurons per layer (f), ray samples per ray (N_{sample}), and noise standard deviation (σ_{std}). These parameters capture the model’s capacity and inductive biases, influencing its ability to reconstruct geometric and radiometric detail.

(ii) Scene-Specific Descriptors. We construct scene-level features capturing view consistency and geometric sampling patterns, both critical for NeRF reconstruction accuracy.

We define valid pairs (i, j) via angular constraints:

$$\mathcal{P} = \{(i, j) \mid i < j, \text{angle}_{i,j} \leq \theta_{\text{view}}, \text{yaw}_{i,j} \leq \theta_{\text{yaw}}\}, \quad (3)$$

where θ_{view} and θ_{yaw} are the pair-wise difference of satellite viewing angle and yaw degree respectively.

The weighted average inverse PSNR w' is defined as:

$$w' = \frac{\sum_{(i,j) \in \mathcal{P}} (d'_i + d'_j) \cdot \text{PSNR}_{ij}^{-1}}{\sum_{(i,j) \in \mathcal{P}} (d'_i + d'_j)}, \quad (4)$$

where $\text{PSNR}_{ij}^{-1} = \frac{1}{\text{PSNR}(I_i, I_j)}$ penalizes dissimilar image pairs, and $d'_i = \max(d_i, \epsilon)$ assigns a minimum degree to images with no similar neighbors. The degree weight d'_i favors images with more neighbors (d_i for image I_i), with $\epsilon > 0$ as minimum contribution.

In addition to the weighted inverse PSNR, we compute the pairwise standard deviation of pairwise inverse PSNR values:

$$\text{std}(\mathcal{P}) = \sqrt{\frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left(\text{PSNR}_{ij}^{-1} - w' \right)^2} \quad (5)$$

To complement the above, we include two additional descriptors. The *pair density* $p(\phi)$ measures the fraction of valid image pairs among all possible view combinations:

$$p(\phi) = \frac{|\mathcal{P}|}{M(M-1)}, \quad (6)$$

where M is the total number of views.

The *coverage score* C quantifies the proportion of input images that participate in at least one valid pair:

$$C = \frac{1}{M} \sum_{i=1}^M c_i, \quad \text{where } c_i = 1 \text{ if } I_i \in \mathcal{P} \quad (7)$$

We compute the *mean cosine similarity* $\overline{\text{cos sim}}$ between view direction vectors to quantify angular alignment across multi-view observations. Higher values indicate strong directional coherence, which benefits volumetric integration.

$$\overline{\text{cos sim}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\| \|\mathbf{d}_j\|}, \quad (8)$$

where $\mathbf{d}_i$ and $\mathbf{d}_j$ are the unit view direction vectors for cameras i and j , and $\mathcal{P}$ is the set of valid image pairs.

The *image variance* $\text{Var}(I_i)$ captures photometric diversity across a scene, including shadows, occlusions, or surface variation that challenge radiometric modeling.

$$\overline{\text{Var}(I_i)} = \frac{1}{M} \sum_{i=1}^M \text{Var}(I_i), \quad (9)$$

where $\text{Var}(I_i)$ is the pixel-level variance of image I_i .

Additionally, we include view-angle statistics (mean, max, std) to capture the overall camera configuration.

Table 1: PSNR statistics across two datasets. The results show diversity in reconstruction difficulty and complexity.

Dataset	PSNR (Avg.)	PSNR (Max)	PSNR (Min)
JAX	19.63	23.03	14.16
OMA	17.18	23.06	12.21

Table 2: Comparison of NAS (on A6000 GPU) and PreSCAN (on Tesla T4, weaker GPU) prediction time. PreSCAN achieves $> 1000\times$ speedup over NAS with comparable accuracy, enabling sub-minute (< 30 sec.) prediction per scene.

(a) Average		(b) Three Example Scenes		
Method	Time	Scene	NAS	PreSCAN
NAS	9.7 hrs	JAX-467	10.1 hrs	28.6s
PreSCAN	28.3s	JAX-258	9.6 hrs	28.9s
		OMA-353	9.9 hrs	28.4s

Prediction Model. We concatenate architectural and scene-level descriptors into a unified feature vector and use a parametric performance estimator to predict reconstruction performance:

$$\text{PSNR}_{\text{pred}} = \beta_{\text{arch}}^T \mathbf{X}_{\text{arch}} + \beta_{\text{scene}}^T \mathbf{X}_{\text{scene}} + \beta_0, \quad (10)$$

where $\mathbf{X}_{\text{arch}}$ represents architectural configurations ($L, f, N_{\text{sample}}, \sigma_{\text{std}}$), and $\mathbf{X}_{\text{scene}}$ denotes scene-level features ($w', \text{std}(p), p(\phi), C, \cos \text{sim}, \text{Var}(I_i)$). The vectors β_{arch} and β_{scene} are the learned weights for each feature type, and β_0 is a scalar bias term.

We deliberately choose a parametric shallow performance estimator over more complex models (e.g., deep neural networks, gradient boosting) for three reasons: (1) our SHAP analysis reveals approximately linear relationships between scene descriptors and reconstruction quality, (2) linear models provide direct interpretability of feature contributions, and (3) they generalize well under limited training data without overfitting. Our evaluation confirms that this simple model achieves < 1 dB prediction error, demonstrating that additional model complexity is unnecessary.

4 Evaluation

4.1 Evaluation Setup

We evaluate PreSCAN on 73 satellite scenes (33 JAX, 40 OMA) from DFC2019 [34] using S-NeRF [8] as the base model, extracting six scene descriptors (discussed in the previous section) per scene. NAS uses an A6000, while PreSCAN runs on a Tesla T4 (a weaker GPU than the A6000).

4.2 PreSCAN Model Construction

Running NAS on S-NeRF across 73 scenes yields 3650 scene-architecture pairs (50 top-performing architectures per scene) for training our prediction model (Eq.10). The NAS search space is defined over architectural parameters known to significantly impact NeRF performance.

- **Number of MLP layers (L):** Chosen from $\{4, 6, 8, 10\}$ to span shallow to deeper networks.

Table 3: Comparison of actual vs. predicted average PSNR for scenes. We report MAE (absolute PSNR gap, $\Delta\text{PSNR} = |\text{Actual} - \text{Predicted}|$). Bold cells indicate low MAE (best predictions), and values marked as (worst) denote high error.

Scene	Actual PSNR	Predict. PSNR	MAE (ΔPSNR)	Std. Error	Bias
JAX-467	20.22	20.74	0.52	0.71	+0.52
JAX-474	20.15	20.57	0.42 (best)	0.31	+0.42
JAX-505	19.14	19.70	0.56	0.74	+0.56
JAX-559	19.06	19.95	0.89 (worst)	0.62	+0.89
OMA-258	19.10	19.04	0.07 (best)	0.13	-0.07
OMA-287	15.88	16.14	0.27	0.21	+0.27
OMA-315	17.12	16.64	0.48	0.17	-0.48
OMA-331	15.59	16.49	0.89 (worst)	0.20	+0.89
OMA-353	16.74	17.26	0.52	0.13	+0.52

- **Hidden layer dimensions (f):** Selected from $\{64, 128, 256\}$ to control model capacity.
- **Samples per ray (N_{sample}):** Drawn from $\{16, 32, 64, 72\}$ to vary volumetric resolution.
- **Sampling noise standard deviation (σ_{std}):** Uniformly sampled from $\{0.0, 0.12, 0.14\}$ for noise robustness.

The hyperparameter combinations yield 144 possible S-NeRF configurations. We use TPE [39] to select the top-50 architectures per scene via NAS, requiring ≈ 10 hours per scene on an A6000 GPU. Table 1 shows PSNR variations across two datasets, confirming scene diversity.

We train on 64 scenes (3200 pairs) using ordinary least squares, which converges in under one minute. Testing on 9 remaining scenes (four JAX, five OMA) demonstrates that PreSCAN achieves $> 1000\times$ speedup over NAS while maintaining accuracy (Table 2).

4.3 Predictive Performance of PSNR Estimation

We evaluate PreSCAN on nine test scenes (4 JAX, 5 OMA) with varying reconstruction complexity. Table 3 reports predicted vs. actual PSNR and error metrics, e.g., each scene’s MAE (ΔPSNR), bias, and standard deviation of errors.

PreSCAN achieves MAE of 0.06 – 0.89 dB in PSNR across nine test scenes, with most predictions within 0.5 dB of ground truth. JAX scenes exhibit slight overprediction (positive bias), while OMA scenes vary more: OMA-258 shows excellent accuracy (0.07 dB MAE), whereas JAX-559 and OMA-331 reach the upper error bound (≈ 0.89 dB). All errors remain practical for deployment and show PreSCAN’s robust and reliable NeRF prediction ability.

Our results reveal a critical insight: multi-view consistency among nearby viewpoints drives PSNR prediction accuracy more than architectural choices. Figure 3 further shows that prediction errors maintains < 1 dB error across all architectures and scenes, with most achieving < 0.5 dB precision. Even challenging scenes (JAX-559, OMA-331) remain within bounds, while well-aligned scenes (OMA-258) achieve 0.07 dB MAE. This validates PreSCAN’s consistent performance despite scene variations.

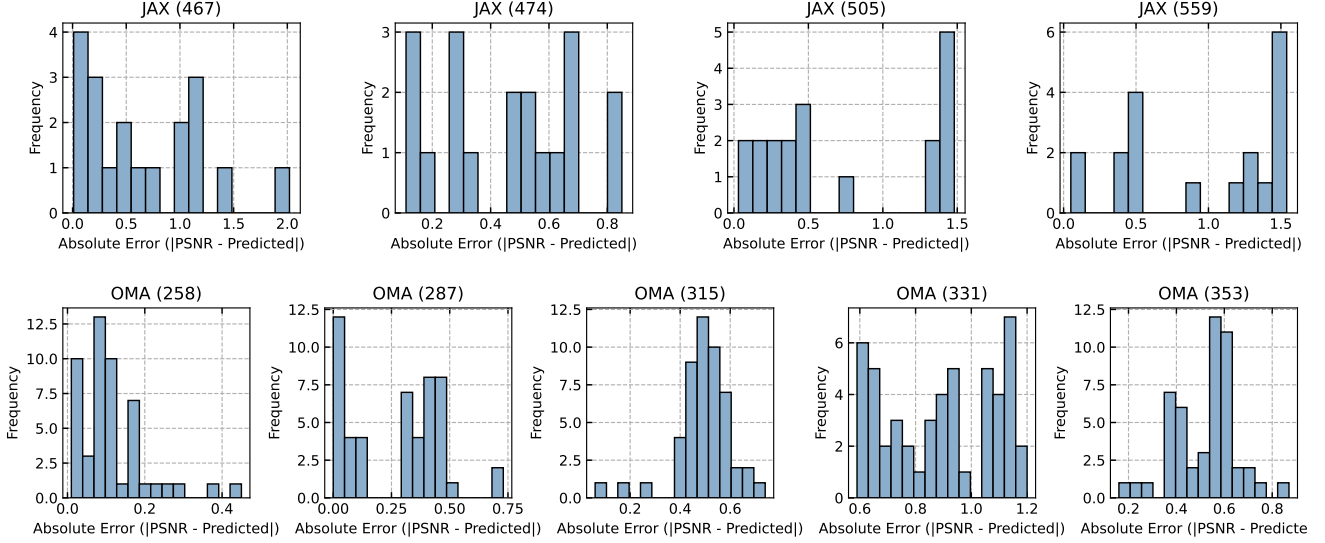


Figure 3: MAE (Δ PSNR) distributions for different NeRF architectures across scenes. Each subplot shows prediction consistency per scene, with most errors < 1 dB and many clustering < 0.5 dB, demonstrating PreSCAN’s prediction accuracy across scenes.

To complement the quantitative results above, Figure 4 shows qualitative comparisons on two representative scenes. On JAX-214 (cityscape) and OMA-331 (suburban), the novel views rendered by the PreSCAN-selected architecture are visually indistinguishable from those produced by both the largest candidate architecture (10 layers, 256 hidden units, 72 samples/ray) and the NAS-selected architecture, despite requiring more than three orders of magnitude less selection time. All three reconstructions preserve building facades, rooftop structures, and cast shadow patterns consistent with the ground truth, confirming that the sub-1 dB PSNR gap reported in Table 3 translates to negligible perceptual difference. Beyond visual quality, the architecture choice has direct implications for training cost: the PreSCAN-selected architecture reduces per-epoch training time from an average of 567.6 s (largest model) and 431.2 s (NAS-selected) to 357.8 s, a 37.0% and 17.0% reduction respectively, with minimal corresponding loss in perceptual quality. This per-epoch saving compounds across the thousands of epochs typical of NeRF training, translating to substantial wall-clock and energy savings at deployment scale.

4.4 Architectural Error Distributions

Beyond average performance metrics, we analyze how architectural parameters affect prediction accuracy by examining Δ PSNR distributions across key design choices, including layers, features, and sampling density (Figure 5).

Figure 5 reveals hidden architectural trends. Models with 128 features show tighter error distributions than 64-feature variants despite similar means, indicating superior robustness. 10-layer networks exhibit wider error spreads, suggesting overfitting, while sampling beyond 32 rays yields diminishing returns. These findings validate that model capacity outweighs sampling density for satellite NeRF performance.

These architectural trends demonstrate PreSCAN’s ability to capture fine-grained performance patterns beyond scene-level statistics. The distinct error profiles for different depths and capacities confirm that PreSCAN captures prediction patterns across both scene and architectural variations in our evaluation, essential for practical NAS replacement. Such predictable error distributions can enable uncertainty- or confidence-aware architecture selection in future deployment scenarios.

4.5 Sparse Input Model Training

To assess generalization under limited supervision, we retrain PreSCAN using only 45 scenes. The *linear relationship between descriptors and PSNR should enable strong performance even in data-scarce settings*, a critical requirement for new geographic regions. Table 4 validates this hypothesis. PreSCAN maintains < 1 dB errors on most unseen scenes, demonstrating robustness in sparse training scenarios.

Results confirm consistent accuracy across the tested scene types. Most predictions achieve excellent accuracy: JAX-314 (0.08 dB) and OMA-142 (0.26 dB) show particularly low errors. Even challenging cases like JAX-359 (0.83 dB) and OMA-244 (1.85 dB, due to extreme viewing geometry) remain practical. Moreover, we observe that prediction errors correlate strongly with actual scene difficulty. *i.e.*, higher errors reliably indicate problematic inputs requiring special handling. This property enables PreSCAN to serve dual roles in operational pipelines: 1) selecting appropriate architectures and 2) flagging scenes for quality review.

4.6 Model Invariance

To test our approach against model invariance, we introduce an additional branch that models architecture-specific effects as $\beta^T X_{\text{model}}$ where X_{model} denotes the model variant used. In our training database, we include S-NeRF and SatNeRF as model choices.

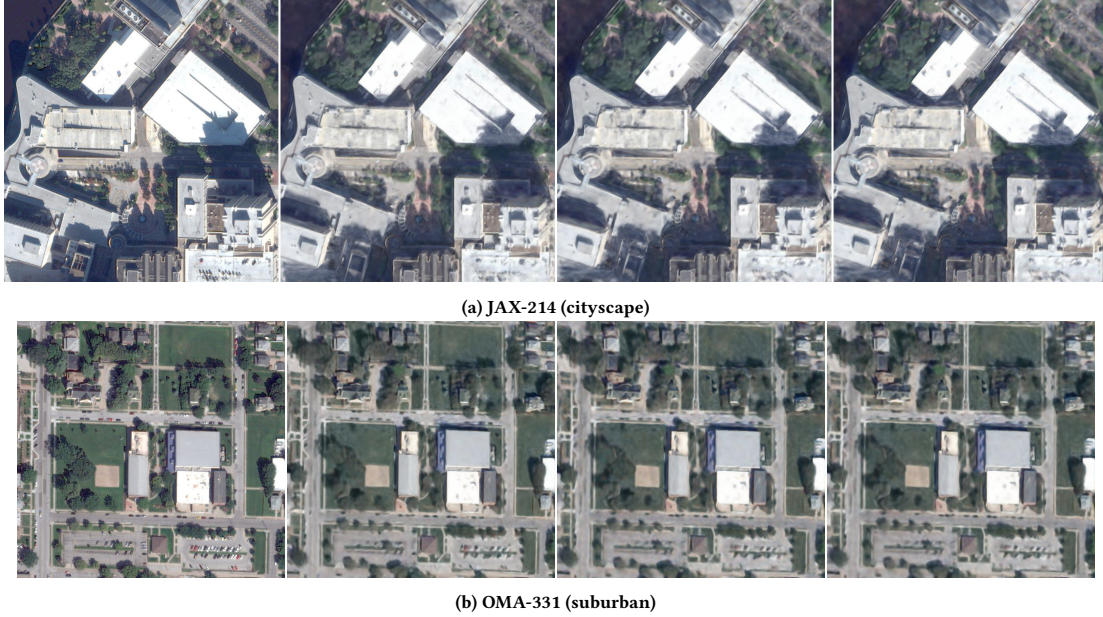


Figure 4: Qualitative reconstruction comparison on representative scenes. Top row: JAX-214 (cityscape); bottom row: OMA-331 (suburban). From left to right: held-out ground-truth view, reconstruction from the largest candidate architecture (10 layers, 256 hidden units, 72 samples/ray), reconstruction from the NAS-selected architecture (9.6+ hours of search), and reconstruction from the PreSCAN-selected architecture (< 30 seconds of selection). All three reconstructions use fully-trained S-NeRF models; only the architecture-selection cost differs. The PreSCAN-selected reconstructions are visually comparable to both the largest model and the NAS-selected ones despite a >1000 $\times$ reduction in selection time.

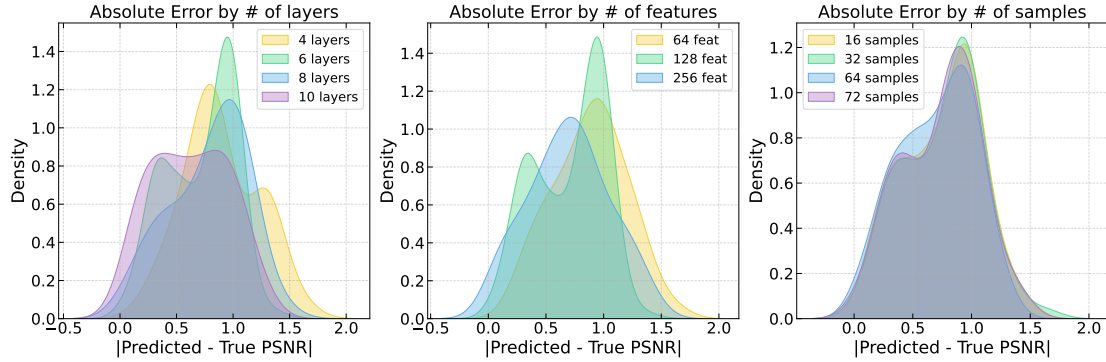


Figure 5: Distribution of absolute prediction errors (Δ PSNR) across various architectural parameters. (a) ‘Layers’: NeRF models with 6 and 8 layers yield tighter, more stable error distributions, while 10-layer models show greater spread and occasional outliers. (b) ‘Feature Dimensionality’: 128-feature models produce the most consistent predictions. (c) ‘Number of Samples’: Increasing the number of samples > 32 offers diminishing returns, with all settings clustering near 1 dB error.

As shown in Table 5, the prediction error remains consistently below 1 dB across both S-NeRF and SatNeRF variants, indicating that our approach captures scene-driven reconstruction difficulty rather than model-specific biases, enabling NeRF variant-agnostic performance estimation.

5 Case Study: Hardware-Aware Deployment

PreSCAN enables fast architecture selection by predicting reconstruction quality prior to training. However, real-world deployment

on satellite onboard systems, UAVs, and edge devices imposes strict power and latency budgets that often preclude selecting the highest-accuracy (typically largest) architecture.

To demonstrate PreSCAN’s practical value in resource-constrained settings, we present a deployment case study on edge platforms. Specifically, we show that PreSCAN’s scene-aware quality predictions can be directly combined with offline hardware cost profiles to enable cost-efficient architecture selection without additional model training or modification.

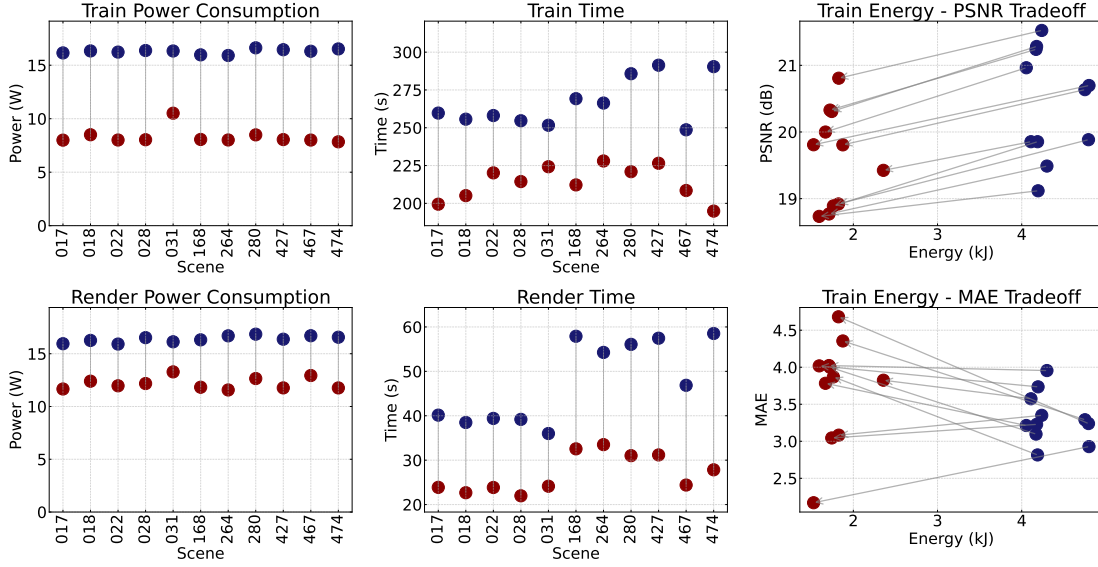


Figure 6: Comparison of power consumption and time between the most accurate model and the cost-efficient model on the Jetson Orin Nano (Blue indicates baseline, red indicates PreSCAN-selected). PreSCAN with hardware-aware architecture selection achieves significant efficiency gains with minimal quality loss.

Table 4: Scene-wise PSNR Generalization Comparison: Sparse Training (45 scenes). Bold cells indicate low MAE (best), and values marked as (worst) denote high MAE error.

JAX (Unseen Scenes)				
Scene	Act. PSNR	Pred. PSNR	MAE	Bias
JAX-276	19.88	19.24	0.64	-0.64
JAX-280	18.95	18.70	0.25	-0.25
JAX-314	18.53	18.60	0.08 (best)	+0.08
JAX-359	17.94	18.77	0.83 (worst)	+0.83
JAX-412	18.46	19.11	0.65	+0.65
JAX-416	19.40	19.28	0.13	-0.13
OMA (Unseen Scenes)				
Scene	Actual	Pred.	MAE	Bias
OMA-142	20.19	19.93	0.26	-0.26
OMA-212	16.54	15.88	0.66	-0.66
OMA-244	19.31	17.46	1.85 (worst)	-1.85
OMA-247	15.55	16.06	0.51	+0.51
OMA-276	15.49	16.84	1.34	+1.34
OMA-278	15.80	16.30	0.50	+0.50
OMA-355	17.25	17.21	0.04	-0.04
OMA-376	18.57	18.22	0.35	-0.35
OMA-389	17.88	17.89	0.01 (best)	+0.01

In this case study, we use two widely-used edge platforms: Nvidia Jetson Orin Nano and AGX. Both devices are built on Nvidia Ampere architecture. The former is a small and more affordable device with lower amount of memory and power consumption, while the latter is a powerful device commonly utilized in advanced applications.

Table 5: Comparison of MAE ($|\text{Actual} - \text{Predicted PSNR}|$) across NeRF variants. We report absolute PSNR error ($|\text{Predicted} - \text{Actual}|$) for S-NeRF and SatNeRF.

Scene	S-NeRF MAE (Δ PSNR)	SatNeRF MAE (Δ PSNR)
JAX-467	0.79	0.82
JAX-474	0.55	0.77
JAX-505	0.61	0.91
JAX-559	0.03	0.57
OMA-258	0.19	0.02
OMA-287	0.37	0.56
OMA-315	0.31	0.41
OMA-331	0.72	0.49
OMA-353	0.88	0.17

Both devices run Ubuntu 22.04 OS and Nvidia JetPack SDK version 6.2. Table 6 summarizes their specifications.

5.1 Offline Deployment Cost Profiling

Deployment costs (latency and power consumption) are primarily driven by architectural parameters (e.g., network depth, width, sampling density) rather than scene content. Unlike reconstruction quality, which varies by scene, the computational cost for a fixed architecture remains consistent. This independence allows us to decouple cost estimation from scene-specific quality prediction.

Leveraging this, we construct a deployment cost database by profiling all candidate NeRF architectures offline on target edge platforms (Jetson Orin Nano [28] and AGX [29]). For each architecture, we measure four key metrics: training power, rendering power,

Table 6: Hardware Specifications of NVIDIA Jetson Devices.

	Orin Nano [28]	Orin AGX [29]
CPU	6-core Cortex-A78AE 1.7GHz	12-core Cortex-A78AE 2.2GHz
GPU	1024 CUDA cores 32 Tensor cores 1.02 GHz	2048 CUDA cores, 64 Tensor cores 1.3 GHz
Memory	Unified 8GB	Unified 64GB
Power Limit	25W	75W

training time, and rendering time. This one-time profiling creates a lookup table that complements PreSCAN’s scene-aware quality predictions, enabling cost-benefit analysis for any new scene.

5.2 Hardware-Aware Architecture Selection

We aim to identify the architecture that minimizes energy and latency while meeting a minimum quality requirement. Given a target device and a quality threshold $PSNR_{\min}$, the selection process follows a simple three-step logic:

- (1) **Predict Quality:** PreSCAN estimates the PSNR for all candidate architectures for the specific scene.
- (2) **Filter:** Architectures failing to meet $PSNR_{\min}$ are discarded.
- (3) **Optimize Cost:** Among the remaining valid candidates, we query the deployment cost database to select the architecture with the lowest energy consumption or latency.

This approach bypasses iterative training or hardware-in-the-loop testing during operation. Please note that this selection pipeline requires no modification to PreSCAN itself; it operates entirely on PreSCAN’s existing quality predictions combined with precomputed cost profiles.

5.3 Empirical Efficiency Gains

We evaluate this hardware-aware strategy by comparing it against an unconstrained baseline (Best Accuracy) that selects the architecture with the highest predicted PSNR. For this evaluation, we applied a strict quality budget relative to the maximum achievable accuracy for a set of test scenes.

Figure 6 visualizes the trade-offs between deployment cost and reconstruction performance. The results indicate that accepting a marginal difference in reconstruction quality allows for the selection of significantly more efficient architectures. As summarized in Table 7, this approach yields substantial reductions in both power consumption and latency for training and rendering, with a negligible impact on the final PSNR. These findings confirm that PreSCAN bridges high-fidelity NeRF reconstruction and the strict resource limits of satellite onboard and edge platforms.

Figure 7 provides a visual counterpart to the trade-off summarized in Table 7. The RGB view and DSM reconstruction from the best-accuracy architecture and from the PreSCAN cost-efficient selection are visually similar, with a minimal difference in color of the rendered RGB views rather than in the overall scene structure observed in the DSM. This suggests that the 0.79 dB PSNR gap corresponds to minor sharpening loss rather than structural

Table 7: Average percentage of benefit from hardware-aware architecture selection with the tradeoff in accuracy.

PSNR Diff	0.79 dB
MAE Diff	0.4 m
Power Consumption – Train	49%
Power Consumption – Render	26%
Time Reduction – Train	19%
Time Reduction – Render	43%

reconstruction failure, consistent with the efficiency trade-off being acceptable for downstream applications that do not require pixel-level photometric fidelity.

6 Deployment Experience and Path Forward

Developing PreSCAN across diverse satellite scenes and edge platforms suggested a set of practical observations, which will be useful for practitioners applying NeRF-based reconstruction at operational scale. We now share these lessons, describe how PreSCAN can be deployed in practice, and outline open challenges.

6.1 Lessons from Building PreSCAN

Building and evaluating PreSCAN across 73 scenes, 10,512 trained configurations, and two edge platforms surfaced several practical lessons that we believe generalize beyond our specific setting.

Lesson 1: Scene quality dominates architectural choice. Our SHAP analysis (Figure 2, §4) shows that scene-level descriptors (inverse PSNR, photometric variance, view-angle cosine similarity) consistently outrank all architectural parameters. Architectural tuning therefore operates within a quality band largely determined by scene-level view consistency, making the marginal return of per-scene NAS small relative to its cost. Practitioners deploying NeRF at scale should prioritize view selection and acquisition-side filtering over architecture search.

Lesson 2: Simple models are all you need when features are well-chosen. A lightweight regression model over 10 features achieves <1 dB MAE and trains in under a minute. The relationship persists under sparse supervision on 45 scenes (Table 4, §4), showing it is not a low-data artifact. For this problem, feature design guided by a one-time SHAP analysis matters more than predictor complexity.

Lesson 3: Prediction error correlates with scene difficulty. Large PreSCAN errors (e.g., OMA-244 at 1.85 dB under sparse training) tend to occur on scenes with extreme viewing geometry or low pair density, which are also challenging for NeRF. Prediction error may therefore serve as a secondary operational signal, flagging scenes for human review, denser acquisition, or fallback reconstruction. Formal calibration is left to future work.

Lesson 4: Cost and quality should be decoupled. Hardware cost (latency, power) is driven largely by architecture, while reconstruction quality depends on both architecture and scene content. This asymmetry allows cost profiles to be measured once per device and reused across scenes, enabling the training and rendering savings in our case study without modifying PreSCAN.

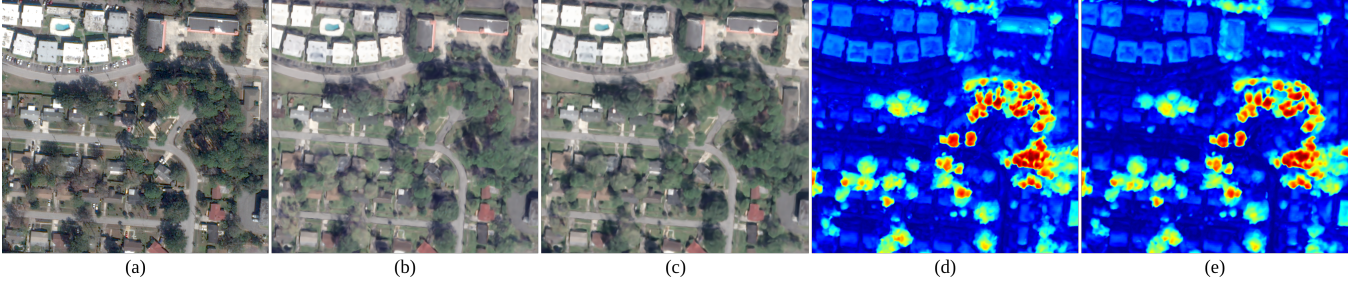


Figure 7: Visual comparison of quality-efficiency trade-off for JAX-427. (a) Ground-truth view. (b) Rendering from the highest-accuracy architecture (baseline). (c) Rendering from the PreSCAN-selected cost-efficient architecture (54.6% lower training power, 60.4% lower training time). (d) DSM from the highest-accuracy architecture. (e) DSM from the PreSCAN-selected cost-efficient architecture. The nearly identical DSMs indicate that the 0.79dB PSNR gap stems from minor photometric differences rather than geometric errors, supporting cost-efficient deployment.

Lesson 5: Variant-agnostic predictors are achievable. A single model-identity term was sufficient to keep MAE under 1 dB across both S-NeRF and SatNeRF (Table 5), suggesting reconstruction difficulty in satellite NeRF is primarily scene-driven and only secondarily model-driven. A single screening predictor can serve an evolving family of satellite NeRF variants rather than being retrained for each new one.

6.2 Path Forward to Deployment

Extension to additional NeRF variants. Our model-invariance experiment (Table 5, §4) shows PreSCAN’s predictions remain accurate across S-NeRF and SatNeRF when a single model-identity term is added to the feature vector. This suggests a natural extension path: incorporating recent satellite NeRF variants such as SpS-NeRF [45] (dense depth supervision for sparse views) and SAT-NGP [2] (hash-grid encoding for fast relightable reconstruction). Each new variant requires only augmenting the training database with NAS results on a modest set of scenes; the same scene-level descriptors transfer directly. We expect variants with different inductive biases to share the scene-driven difficulty pattern observed here, since the underlying multi-view consistency challenges in satellite imagery are model-agnostic.

Integration with existing remote sensing workflows. PreSCAN is designed to integrate into operational satellite mapping pipelines as a lightweight triage stage between image acquisition and reconstruction. After standard pre-processing (RPC refinement, cloud masking), PreSCAN consumes the view set and its metadata to (i) estimate reconstruction quality for candidate architectures in < 30 seconds, (ii) select an architecture meeting downstream quality and resource budgets, and (iii) optionally flag scenes whose descriptors fall outside the training distribution. Because PreSCAN requires only view geometry and pairwise image statistics (no rendering, no training), it is compatible with both ground-based processing centers and onboard satellite or UAV pipelines where compute is scarce. For drone and aerial photogrammetry, the same descriptors apply with minor adjustments to the angular thresholds in Eq. 3 for narrower baselines. Integration as a pre-reconstruction screening step would let operators allocate expensive NeRF training only to scenes where it is likely to succeed.

6.3 Open Challenges

Our evaluation is based on the DFC2019 benchmark, which samples suburban and industrial North American environments. Generalizing PreSCAN to other scene types will likely require expanding the training database with NAS results from representative scenes in each category, since our descriptors capture view-consistency statistics but not biome-specific reconstruction challenges (e.g., canopy occlusion, specular water surfaces, low-texture sand or snow). A complementary direction is cross-sensor generalization: extending beyond DFC2019’s WorldView-3 imagery to other sensors (e.g., Pléiades, SkySat, PlanetScope) with different ground sample distances and radiometric characteristics. We view both as natural follow-ups rather than fundamental limitations, given the model-agnostic structure of PreSCAN’s feature design.

7 Conclusion

We present PreSCAN, a scene-aware NeRF architecture selection framework that predicts reconstruction quality without full model training or NAS. By combining architectural and photometric-geometric descriptors, PreSCAN enables $1000\times$ faster architecture screening with < 1 dB MAE across diverse satellite scenes. SHAP analysis shows that view consistency is more predictive than architectural complexity, and the model remains accurate under sparse supervision. We further demonstrate PreSCAN’s practical applicability through a hardware-aware deployment case study on edge platforms, showing that its predictions can be combined with offline cost profiling to achieve significant reductions in power and latency with minimal quality loss.

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